Solving Quadratic Equations

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KEY WORDS quadratic *(pūrua)* · solve *(whakaoti)* · solution *(otinga)* · square root *(pūtakerua)*

A Using square roots ●○○

Remember: A positive number has two square roots: $x^2 = 25$ gives $x = 5$ or $x = -5$.

1. Solve. Give both solutions.

a) $x^2 = 49$ _____ **b)** $x^2 = 100$ _____ **c)** $2x^2 = 50$ _____

d) $x^2 + 3 = 28$ _____ **e)** $x^2 - 20 = 16$ _____ **f)** $3x^2 = 48$ _____

g) $2x^2 - 5 = 13$ _____ **h)** $5x^2 + 7 = 27$ _____ **i)** $(x + 1)^2 = 16$ _____

B The zero product property ●●○

Remember: If two things multiply to make 0, then one of them must be 0.

2. Solve.

a) $(x - 3)(x + 5) = 0$ **b)** $(x + 2)(x + 7) = 0$ **c)** $(x - 6)(x - 1) = 0$

d) $x(x - 4) = 0$ **e)** $x(x + 9) = 0$ **f)** $(2x - 1)(x + 3) = 0$

g) $(x - 4)(3x + 2) = 0$ **h)** $3x(x - 2) = 0$ **i)** $(2x + 5)(2x - 5) = 0$

j) $(x + 8)(x + 8) = 0$ **k)** $5x(2x - 1) = 0$ **l)** $(3x - 1)(4x - 3) = 0$

C No number on the end ●●○

Watch out: Do not divide both sides by x. You would lose the solution $x = 0$.

3. Take out the common factor, then solve.

a) $x^2 - 5x = 0$ **b)** $x^2 + 8x = 0$ **c)** $3x^2 + 12x = 0$

d) $2x^2 - 14x = 0$ **e)** $5x^2 + 10x = 0$ **f)** $4x^2 - 12x = 0$

g) $x^2 = 6x$ **h)** $2x^2 = 6x$ **i)** $3x^2 = -15x$

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D Solving x² + bx + c = 0 ●●○

Remember: Factorise into two brackets. Then use the zero product property.

4. Solve.

a) $x^2 + 5x + 6 = 0$ **b)** $x^2 + 8x + 7 = 0$ **c)** $x^2 - 7x + 10 = 0$

d) $x^2 - 10x + 24 = 0$ **e)** $x^2 + 3x - 10 = 0$ **f)** $x^2 - 4x - 21 = 0$

g) $x^2 + 7x - 8 = 0$ **h)** $x^2 - 7x - 18 = 0$ **i)** $x^2 + 8x + 16 = 0$

j) $x^2 - 6x + 9 = 0$ **k)** $x^2 + 7x - 30 = 0$ **l)** $x^2 - 7x - 60 = 0$

E Rearrange first ●●●

Watch out: One side must be 0 before you factorise.

5. Solve.

a) $x^2 + 3x = 10$ **b)** $x^2 = 7x - 12$ **c)** $x(x + 2) = 15$

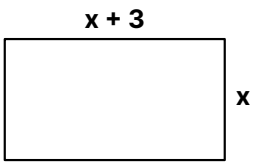
d) $x^2 + 5x + 4 = 10$ **e)** $x^2 - 20 = x$ **f)** $x^2 + 16 = 8x$

g) $x(x - 3) = 10$ **h)** $2x^2 + 3x = x^2 + 18$ **i)** $(x + 2)(x + 3) = 2$

j) $x^2 = 36 - 5x$ **k)** $3x^2 - 7x = 2x^2 + 8$ **l)** $(x - 1)^2 = 2x + 6$

F Problems ●●●

6. The rectangle has an area of 40 cm². Write an equation and solve it to find x.



7. Two consecutive positive whole numbers multiply to make 72. Let the smaller number be x. Find the two numbers.

8. The square of a positive number is equal to 5 times the number, plus 14. Find the number.

★ Extension

9. These have a number in front of x^2. Factorise, then solve.

a) $2x^2 + 7x + 3 = 0$ **b)** $3x^2 - 10x + 8 = 0$ **c)** $6x^2 + x - 2 = 0$

d) $5x^2 - 3x - 2 = 0$ **e)** $4x^2 + 12x + 9 = 0$ **f)** $12x^2 + 17x - 5 = 0$

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Solving Quadratic Equations

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KEY WORDS quadratic *(pūrua)* · solve *(whakaoti)* · solution *(otinga)* · square root *(pūtakerua)*

A Using square roots ●○○

Remember: A positive number has two square roots: $x^2 = 25$ gives $x = 5$ or $x = -5$.

1. Solve. Give both solutions.

a) $x^2 = 64$ _____ **b)** $x^2 = 121$ _____ **c)** $2x^2 = 72$ _____

d) $x^2 + 5 = 21$ _____ **e)** $x^2 - 13 = 36$ _____ **f)** $4x^2 = 36$ _____

g) $2x^2 - 9 = 41$ _____ **h)** $3x^2 + 8 = 20$ _____ **i)** $(x + 2)^2 = 25$ _____

B The zero product property ●●○

Remember: If two things multiply to make 0, then one of them must be 0.

2. Solve.

a) $(x - 4)(x + 6) = 0$ **b)** $(x + 3)(x + 8) = 0$ **c)** $(x - 5)(x - 2) = 0$

d) $x(x - 7) = 0$ **e)** $x(x + 6) = 0$ **f)** $(2x - 3)(x + 4) = 0$

g) $(x - 2)(4x + 1) = 0$ **h)** $2x(x - 5) = 0$ **i)** $(2x + 3)(2x - 3) = 0$

j) $(x + 9)(x + 9) = 0$ **k)** $4x(2x - 3) = 0$ **l)** $(3x - 2)(5x - 1) = 0$

C No number on the end ●●○

Watch out: Do not divide both sides by x. You would lose the solution $x = 0$.

3. Take out the common factor, then solve.

a) $x^2 - 7x = 0$ **b)** $x^2 + 6x = 0$ **c)** $2x^2 + 10x = 0$

d) $3x^2 - 12x = 0$ **e)** $5x^2 + 15x = 0$ **f)** $4x^2 - 20x = 0$

g) $x^2 = 9x$ **h)** $4x^2 = 8x$ **i)** $2x^2 = -14x$

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D Solving x² + bx + c = 0 ●●○

Remember: Factorise into two brackets. Then use the zero product property.

4. Solve.

a) $x^2 + 5x + 4 = 0$ **b)** $x^2 + 8x + 12 = 0$ **c)** $x^2 - 8x + 15 = 0$

d) $x^2 - 10x + 16 = 0$ **e)** $x^2 + 4x - 12 = 0$ **f)** $x^2 - 5x - 24 = 0$

g) $x^2 + 6x - 7 = 0$ **h)** $x^2 - 8x - 20 = 0$ **i)** $x^2 + 10x + 25 = 0$

j) $x^2 - 4x + 4 = 0$ **k)** $x^2 + 5x - 36 = 0$ **l)** $x^2 - 5x - 66 = 0$

E Rearrange first ●●●

Watch out: One side must be 0 before you factorise.

5. Solve.

a) $x^2 + 4x = 12$ **b)** $x^2 = 9x - 20$ **c)** $x(x + 3) = 28$

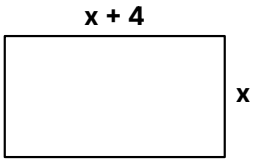
d) $x^2 + 6x + 2 = 9$ **e)** $x^2 - 30 = x$ **f)** $x^2 + 25 = 10x$

g) $x(x - 4) = 12$ **h)** $2x^2 + 5x = x^2 + 14$ **i)** $(x + 1)(x + 4) = 4$

j) $x^2 = 40 - 3x$ **k)** $3x^2 - 5x = 2x^2 + 14$ **l)** $(x - 2)^2 = x + 10$

F Problems ●●●

6. The rectangle has an area of 60 cm². Write an equation and solve it to find x.



7. Two consecutive positive whole numbers multiply to make 132. Let the smaller number be x. Find the two numbers.

8. The square of a positive number is equal to 3 times the number, plus 28. Find the number.

★ Extension

9. These have a number in front of x^2. Factorise, then solve.

a) $2x^2 + 11x + 5 = 0$ **b)** $3x^2 - 14x + 8 = 0$ **c)** $6x^2 - 7x - 3 = 0$

d) $5x^2 - 7x - 6 = 0$ **e)** $4x^2 + 20x + 25 = 0$ **f)** $12x^2 - x - 6 = 0$

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